Solution Requirements

Date	07 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviours, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviours (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the Credit Card Approval Prediction web application to enter applicant details and view prediction results. (If your application has no login, mention: "No user authentication is required for this prototype application.")	Low

2	Authorization levels	The prototype has a single user role. All users have permission to enter applicant details and view prediction results.	Low
3	External interfaces	The system interacts with the trained Logistic Regression model (best_model.pkl) and the saved label encoders (label_encoders.pkl) for prediction.	High
4	Transactions processing	The system accepts applicant information, preprocesses the data, predicts the approval status using the ML model, and displays the result instantly. No financial transaction or payment processing is involved.	High
5	Reporting	The application displays the prediction result (Approved/Rejected). The trained model's performance metrics (Accuracy: 98.31%) are available in the notebook for evaluation.	Medium
6	Business rules, etc.	The system predicts credit card approval based on applicant demographic, employment, income, education, family, housing, and occupation details using the trained Logistic Regression model.	High

7	Compliance to laws or regulations	The prototype is intended to comply with basic data privacy principles by using applicant information only for prediction purposes and not storing personal data.	Medium
8	*Other*	The Flask web application provides a simple, user-friendly interface for entering applicant information and obtaining instant predictions.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	The system should process user input and return prediction results quickly.	Prediction displayed in **less than 2 seconds**.
2	Scalability	The application should support future enhancements, including deployment for multiple users.	Supports multiple prediction requests without affecting performance.
3	Security & Data Privacy	Applicant information should be used only for prediction and should not be stored permanently.	No sensitive user data is stored after prediction.
4	Reliability & Availability	The system should provide consistent predictions whenever the application is running.	Application availability of **99%** during normal operation.

5	Usability & Accessibility	The web interface should be simple, intuitive, and easy for users to operate.	Users can complete a prediction in **less than 2 minutes** without assistance.
6	*Other*	The application should be maintainable and easy to update with new machine learning models in the future.	Model replacement should require minimal code changes.